

A Machine-Checked Companion to Goertzel’s Spencer–Brown/Kauffman Route to the Four-Color Theorem: Keystone, Pillars, and the Composition Ladder

The MeTTapedia formalization effort

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Abstract

This is a living formalization companion to Ben Goertzel’s manuscript *A Human-Readable, Spencer–Brown/Kauffman-Style Proof of the Four-Color Theorem* (v23), which organizes a candidate human-readable proof of the Four-Color Theorem (4CT) around three interacting pillars: (A) an annular difference-*spanning* statement for a zero-boundary space $W_0(H)$ (Theorem 4.9); (B) a CAP5-free normal form with radius-1 collar locality; and (C) an *executability* layer that, refusing a false additivity of Kempe switches, reduces between-region Kempe connectivity to a single *Local Kempe Reachability with fixed input* (LKR_{in}) property for one canonical three-cell collar gadget τ (Lemma 8.14), composes it along the collar chain (Lemmas 8.15, 8.18, Cor. 8.19), and assembles the wide annulus collar-by-collar (§9.2). We record what the MeTTapedia formalization has machine-checked of this route, with an honest ledger separating what holds outright, what holds modulo one named remainder, and what is open, and we foreground the places where the formalization *refutes* a step rather than discharging it.

Machine-checked outright (no `sorry`; axioms $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$): finite encoded-model audits of the keystone LKR_{in} property (Lemma 8.14) and two-color *transparency* (Lemma 8.15), together with completeness of the 192-state incidence enumeration — graph-level Kempe-step soundness remains open; a proper-subspace strictness theorem against a larger unrestricted boundary-zero chain space, with four uniform peel oracles refuted and conditional need-based span infrastructure — v23’s actual cycle-space inclusion (Theorem 4.9) remains open; and a length-2/3 radial-closing-edge fiber census for §9.2 Step 3 — the manuscript’s all-outer-stub, arbitrary-length statement remains open. The Gate-2 composition (Lemmas 8.18/8.19) is *reduced* — the manuscript’s own pointwise Step-1 preparation is refuted and replaced by a frontier-automaton / fibration architecture — to one live pin combining representative real-fiber connectivity audits with DFA-mode sufficiency.

The formalization also produced a decisive negative in the encoded model. §9.2 Step 4 — the manuscript’s reinterpretation of collar Kempe words as words of the ambient annulus — fails in the encoded two-layer tau-chain model: a certified, collar-input-avoiding switch whose ambient component escapes the inner boundary, re-enters, and changes the collar restriction. Its realization as a simple planar, cyclically-five-edge-connected actual collar is a named open obligation, so this is a regression countermodel against the composition *seam* as written, not a graph-level refutation — and not a refutation of 4CT; its salvage (a normal-form confinement argument, or a face-collar redesign) is open. We make no claim about the Four-Color Theorem — which is of course a theorem, via the Appel–Haken and Robertson–Sanders–Seymour–Thomas computer-assisted proofs. The human-readable v23 route remains *open*, with its foundational layer audited in an encoded model, and its most dangerous assembly seam pinned down as broken in that model.

A third development round turned to the v24 working material (*Deserts, Pivots, and the Necklace*), which replaces v23’s §6–7. The v24 validation toolchain and its numerical claims were reproduced independently: the 5412 and 31704 annular censuses with their Kempe fibers,

the fullerene bases, the pentagon seals, the desert and strand specimens — including the 688-edge strand refuting the package’s own Short Strand Conjecture — and the GWCO exact base through twelve. On the structural side the development now carries a rotation-system Tait-coloring layer with exact facial linear algebra; the all-face discrete Gauss–Bonnet identity (`totalFaceCurvature_eq_twelve`); a proved *weighted* pentagon budget correcting the package’s at-most-twelve pentagon count; the locked-pentagon necklace assembly with singleton-strand forcing; a hexagonal corridor theory (rung classification, clean width-four slabs, oriented interfaces, Tait transfer, a sharp chirality barrier); congruence-correct corridor and tube-ring pumping; and a framed-trail layer with square/digon/rotor fiber calculi. The route’s endgame — fusion chains, chiral monodromy, Trail Completeness with its L1–L10 dependency package, and a proved Short-Strand replacement — remains open; the corridor and pumping infrastructure is the active replacement search.

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1 The problem and Ben Goertzel’s three-pillar route

1.1 Tait’s reduction and the minimal-counterexample target

The Four-Color Theorem asserts that every plane triangulation G admits a proper vertex 4-coloring. Passing to the cubic dual $K = G^*$, Tait’s classical equivalence (v23, Thm. 2.3) restates 4CT as: every bridgeless planar cubic graph admits a *proper 3-edge-coloring*, i.e. a map $\chi : E(K) \rightarrow \{r, b, p\}$ assigning distinct colors to the three edges at each vertex. The manuscript encodes the three colors

as the three nonzero elements of $\mathbb{G} = \mathbb{F}_2^2$, with $r = (1, 0)$, $b = (0, 1)$, $p = (1, 1)$, so that “properness at a vertex” is exactly $\chi(e_1) + \chi(e_2) + \chi(e_3) = r + b + p = 0$ in $\mathbb{G}$, and “integrating” an edge-coloring across the planar dual becomes a path-sum in $\mathbb{F}_2^2$.

Following Spencer–Brown and Kauffman, the manuscript reformulates colorings as plane *formations* (red/blue Jordan curves with purple overlaps), defines Kempe switches as the standard two-color interchange, and proves the *Parity Lemma* (planar invariance of the mod-2 count of two-color circuits). The role of §3 is to reduce 4CT, via Kauffman’s parity/primalty argument (Theorem 3.1), to the nonexistence of a minimal counterexample containing a *between-region annulus* H whose boundary-compatible colorings are *not* connected by Kempe moves supported inside the annulus. Thus the entire analytic burden is shifted to a *between-region Kempe connectivity* statement for an annulus $H \subseteq K$ with outer and inner boundary cycles Γ_∞, Γ_0 .

1.2 The three pillars

The manuscript’s conceptual overview (v23, §1.1) states the argument as three interacting pillars.

Pillar A: annular difference spanning (the linear “accounting” layer). Working in a between-region annulus H , one encodes boundary colorings and their differences as vectors in an $\mathbb{F}_2$ -space. Let $W(H)$ be the coordinatewise cycle space of $\mathbb{G}$ -chains on $E(H)$ satisfying the Kirchhoff/even-degree constraints at interior vertices, and let

$$W_0(H) := \{x \in W(H) : x(e) = 0 \text{ for all boundary edges } e \in B_1 \cup B_2\}$$

be the *zero-boundary* subspace. For a fixed proper coloring C_0 of H with Kempe closure $\text{Cl}(C_0)$, the manuscript defines a family $\text{KC}(C_0)$ of *Kempe-closure (polarized face) generators* $X_f^{ab}(C)$ supported in depth-1 collars (v23, Def. 4.8), and proves:

Theorem 1 (Spanning theorem for $W_0(H)$, v23 Theorem 4.9). *Let H be an annular subgraph of a bridgeless planar cubic graph, with boundary cycles Γ_0, Γ_∞ . Fix any proper 3-edge-coloring C_0 of H and let $\mathcal{G} := \text{KC}(C_0)$. Then $W_0(H) \subseteq \text{span}_{\mathbb{F}_2}(\mathcal{G})$; equivalently, if $z \in W_0(H)$ satisfies $z \cdot g = 0$ for every $g \in \mathcal{G}$, then $z = 0$.*

The manuscript proves Theorem 1 in strong-dual annihilator form: it shows $U^\perp \cap W_0(H) = \{0\}$ for $U = \text{span}_{\mathbb{F}_2}(\mathcal{G})$, by a planar orthogonality peeling along a dual spanning forest, after a “purification” step (Lemma 4.4) that removes the polarization of the face generators.

Pillar B: CAP5-free geometry and radius-1 locality (the geometric locality layer). Assuming a minimal counterexample, the manuscript eliminates the CAP5 separating obstruction and works in a CAP5-free normal form (Corollary 6.9). A tube-forcing argument then reduces every “escape” of an anchored Kempe chain to a finite three-cell gate enumeration, yielding radius-1 confinement of anchored Kempe cycles together with a cap-aware disjointness/commutativity package (Lemmas 7.8, 7.9, 7.10). This is what licenses peeling the wide annulus into radius-1 collars.

Pillar C: executability without false additivity (the algebra-to-geometry bridge). The decomposition $\Delta = \sum v_{f,g}$ provided by Pillar A is a statement in $W_0(H)$; it does *not* by itself imply that one may *execute* the corresponding mixed color-pair Kempe switches as if they formed an abelian sum. The manuscript is explicit that cross-pair interference is real: an (a, b) -switch can destroy the (b, c) -structure a later switch needs (v23, §8.1), and Kempe switches using different color

pairs on shared support need not commute (v23, §8.3). To execute the decomposition honestly, the manuscript moves to an operator model: proper colorings form an orthonormal basis of a Penrose–Kauffman state space, and each Kempe switch is an involutive linear operator (generally noncommuting with switches of other color pairs). Using radius-1 locality, between-region Kempe connectivity is reduced to a single local property:

Definition 1 (Local Kempe reachability with fixed input, v23 Def. 8.12). Let X be a planar cubic tangle with ordered input stubs S_{in} and output stubs S_{out} , and fix an input coloring c . Let $\Omega_X(c)$ be the set of proper 3-edge-colorings of X extending c on the input stubs. The *Kempe graph* $K_X(c)$ has vertex set $\Omega_X(c)$, with an edge whenever two colorings differ by a single Kempe switch supported on a two-color component touching no input-stub edge. We say X satisfies LKR_{in} if $K_X(c)$ is connected for every c for which $\Omega_X(c) \neq \emptyset$.

The chain of Pillar C is then:

- **Lemma 8.14:** the canonical three-cell collar gadget τ satisfies LKR_{in} (and so does its mirror);
- **Lemma 8.15:** two-color *transparency* of τ — every two-color component is a boundary-to-boundary path with no internal cycle;
- **Lemma 8.18 / Corollary 8.19:** LKR_{in} *composes* across a transparent interface, so an entire cut-open radius-1 collar tangle $\hat{A} = G_m \circ \cdots \circ G_1$ satisfies LKR_{in} ;
- **§9.2 assembly:** peel the wide annulus H into nested radius-1 collars and use the collar LKR_{in} property to rewrite the current coloring to the target collar-by-collar from outside to inside, the cap-aware locality lemma (7.9) ensuring no switched component reaches back to change a fixed boundary.

1.3 Why Lemma 8.14 is the keystone

The manuscript states that its proof is human-readable and “the only finite verification required is a small by-hand audit of a single three-cell gadget” (v23, Abstract and §1). In the three-pillar architecture, every other step is either an explicit local lemma or a structural reduction; the one place where a finite, exhaustive check is genuinely unavoidable is the LKR_{in} property of the canonical gadget τ . Formalizing Lemma 8.14 therefore certifies exactly the finite core that the manuscript isolates as load-bearing. The remainder of this companion describes that formalization (§2), the resolution of the Pillar A spanning question into a need-based repair (§3), the Pillar C composition ladder — what is proved, what is reduced to one pin, and where a decisive counterexample was found (§4) — and a precise status ledger (§6). Superseded routes are recorded in appendices.

2 The Pillar-C keystone, machine-checked: Lemma 8.14 and transparency 8.15

We restate the manuscript’s keystone in its own notation and then describe the Lean encoding.

Lemma 1 (LKR_{in} for the canonical three-cell gadget, v23 Lemma 8.14). *Let τ be the canonical three-cell collar gadget. It is a planar cubic tangle with internal vertices $F_0, \dots, F_5$ and boundary stubs $B_0, \dots, B_7$, whose internal edges form the tree*

$$F_1 - F_0 - F_2 - F_4 - F_5, \quad \text{with a leaf } F_3 \text{ attached to } F_2,$$

and whose boundary stubs are attached by $F_0B_0, F_1B_1, F_1B_2, F_3B_3, F_3B_4, F_5B_5, F_5B_6, F_4B_7$. Take input stubs $\{B_0, B_1, B_2, B_3\}$ and output stubs $\{B_4, B_5, B_6, B_7\}$. Then τ satisfies LKR_{in} . The same holds for the mirror gadget.

The manuscript proves Lemma 1 by cutting τ along the internal interface edge $I = F_2F_4$ into $\tau = \sigma_R \circ \sigma_L$ and doing a small case analysis on the two pieces. Our formalization instead audits the *full* gadget directly, by exhibiting the proper-coloring state space and an explicit Kempe-path certificate connecting every fixed-input fiber; this is a strictly more direct realization of the same LKR_{in} claim, and it does not rely on the σ_L, σ_R split or on the composition Lemma 8.18.

2.1 The canonical gadget and its 192-state coloring space

The Lean module fixes the colors and the gadget incidence exactly as in Lemma 1: the three nonzero colors are an inductive type, and the gadget's edges (five internal edges plus the eight boundary stubs) and internal vertices are inductive types with an explicit incidence map.

```
inductive LColor | r | b | p
  deriving DecidableEq, BEq, Repr, Inhabited

inductive TauEdge
  | F1F0 | F0F2 | F2F3 | F2F4 | F4F5
  | B0 | B1 | B2 | B3 | B4 | B5 | B6 | B7
  deriving DecidableEq, BEq, Repr, Inhabited

def tauInputEdges : List TauEdge := [TauEdge.B0, TauEdge.B1, TauEdge.B2, TauEdge.B3]
```

A proper 3-edge-coloring of τ is represented by the finite choices left after properness is imposed at every cubic vertex: an ordered color triple at the central vertex F_2 , one binary choice for the forced edge F_1F_0 , and one ordered-pair (swap) bit at each remaining boundary fork — $6 \times 2^5 = 192$ states. That this enumeration is exactly the proper-coloring state space is a checked fact (192 entries, no duplicates, proper at every vertex).

2.2 The Kempe audit and the machine-checked theorems

A Kempe step switches a two-color component *disjoint from the input stubs* (Definition 1). The LKR_{in} property is witnessed by an explicit indexed-path certificate: for each of the 192 states it records the representative of that state's fixed-input fiber and a path of at most two input-disjoint Kempe switches reaching it, plus a coherence audit that any two states with the same input edges share a representative (so fibers are well defined). The three finite audits each reduce to **true** by the kernel's decision procedure; the only axiom dependency is **propext**.

```
theorem tauStateSpaceAudit_ok : tauStateSpaceAudit = true := by decide
theorem lemma814PathCertificate_rows_ok :
  validIndexedRowsFrom 0 lemma814PathCertificate = true := by decide
theorem representativeCoherenceAudit_ok : representativeCoherenceAudit = true := by
  decide

/-- Lemma 8.14, closed by the finite state-space and indexed path certificate. -/
theorem lemma814_tau_lkrIn_audit : lemma814_tau_lkrIn_finiteCheck = true := by
  simp [lemma814_tau_lkrIn_finiteCheck, tauStateSpaceAudit_ok, tauLKRInAudit,
    lemma814PathCertificateAudit_ok]
```

The mirror clause holds definitionally (reflection changes only the planar order of an isomorphic incidence tree). The incidence enumeration is complete for the encoded tau graph: every proper incidence coloring is one of the 192 enumerated colorings. The finite certificate audit is proved, but graph-level soundness of every encoded step remains the named obligation `Lemma814EncodedMovesAreGraphKempeSwitchesTarget`. Thus the current Lemma 8.14 result is a finite encoded-model audit, not yet the manuscript graph theorem.

Remark 1 (Transparency 8.15, finite encoded-model audit). *The same module formalizes Lemma 8.15 (two-color transparency): for each of the 192 colorings, every nonempty two-color component is a boundary-to-boundary tree path (`transparentComponent`), aggregated as `tauTreeTransparencyAudit`. That audit is now proved — `lemma815_tau_tree_transparency_audit` reduces to `true` by a range decomposition of the finite check, axiom-clean. The encoded transparency audit is proved. Its use as graph-level composition (§4) shares Lemma 8.14’s named graph-fidelity obligation, so the local graph-theoretic core is not yet sealed.*

3 Pillar A: a proper-subspace obstruction, and its need-based repair

Pillar A (the spanning Theorem 1) is where the formalization first found — and then routed around — a real gap. We give both halves, because together they are a clean example of what a companion of this kind is for.

3.1 The obstruction: the spanning *equality* over-reaches

On every *closed-walk exact shell* — a concrete annular configuration carrying the boundary-source and interior-endpoint witness data that Theorem 1 is meant to cover — the boundary-zero target that the Kempe-closure detector actually pins down is a *strict* subspace of the full boundary-zero chain space.

```
-- On every closed-walk exact shell, the Theorem-4.9 boundary-zero target is a
proper subspace of the full selected-boundary-zero chain space. -/
theorem
  theorem49BoundaryZeroKirchhoffSubspace_lt_planarBoundaryZeroSubmodule_of_closedWalkExactShell

  [Fintype G.edgeSet] {emb : PlaneEmbeddingWithBoundary G}
  (shell : ClosedWalkExactShell emb) :
  theorem49BoundaryZeroKirchhoffSubspace emb
    (selectedBoundaryInteriorEdgeEndpointVertices emb) <
    planarBoundaryZeroSubmodule emb
```

The strict inequality `<` is the load-bearing content: a proved *proper*-subspace inclusion, not a heuristic. The structural reason (every closed-walk shell forces two distinct interior-support edges onto one face boundary) refutes the natural one-collar repairs, and a companion audit shows the four proposed *uniform geometric oracles* that would each close Theorem 1’s synthesis are all false:

```
theorem not_interiorDualPeelOracle      : ¬ InteriorDualPeelOracle
theorem not_wellFoundedPeelOracle       : ¬ WellFoundedPeelOracle
theorem not_previousBoundaryGeometryOracle : ¬ PreviousBoundaryGeometryOracle
theorem not_residualBoundaryGeometryOracle : ¬ ResidualBoundaryGeometryOracle
```

Lean proves that the formal detector is strictly smaller than a larger unrestricted selected-boundary-zero chain space. This does not refute v23’s actual cycle-space/Kirchhoff-constrained claim $W_0(H) \subseteq \text{span}_{\mathbb{F}_2}(\mathcal{G})$. That inclusion is now named `V23Theorem49W0ContainedInKempeClosureGeneratorSpanTarget` and remains open; the four uniform oracle routes that would have closed it uniformly are refuted.

3.2 The repair: the route never needs the full space

The resolution is that the synthesis does not *use* the full boundary-zero space; it uses only the discrepancy classes of boundary-compatible coloring pairs, and those lie in the detector subspace the manuscript’s own peeling witnesses span. Formally, the requirement actually consumed downstream is *need-based*: it suffices that (1) the detector subspace is contained in the projected Kempe-closure generator span, (2) each execution step preserves that subspace, and (3) every route-used discrepancy class lies in it. All three are machine-checked in the route’s natural cubic setting (module `PillarASmallerSpace`):

```
-- (3) discrepancy classes lie in the detector subspace (cubic setting):
theorem routeUsedDiscrepancyClassesLieInDetectorSubspace_of_hasCubicIncidentEdgeTriples
  ...
-- (2) a Kempe-switch execution step preserves detector membership:
theorem routeKempeSwitchExecutionStep_preserves_detectorSubspace ...
-- assembled sufficiency (inclusion form), discharging the requirement:
theorem
  needBasedPillarARequirement_of_detector_le_span_of_stepPreserves_of_routeDiscrepancies
  ...
```

Proposition (3) has a two-idea proof one can state aloud: a Tait coloring is a Klein-four flow at each cubic vertex, so two colorings agreeing on the selected boundary differ by a *boundary-zero flow*, which is exactly what the Kirchhoff detector certifies. The manuscript’s Theorem 4.9 states an *inclusion*, not an equality. The existing need-based theorems are conditional on detector containment, step preservation, discrepancy membership, and embedding-specific peel data. They are conditional algebraic-span infrastructure, not a proof of Theorem 4.9. What remains a genuine per-embedding obligation is the *supply* of a peeling witness for the synthesis’s actual embedding class; that is a composition-stage task, not a spanning gap.

4 Pillar C: the composition ladder

Between the local keystone (§2) and 4CT, v23’s own argument climbs a ladder: single-gadget LKR_{in} → transparent composition to a whole collar chain (Gate-2) → closed re-glued collars (§9.2 Step 3) → execution in the ambient annulus (§9.2 Step 4) → the wide-annulus peel. This companion has now walked a substantial part of that ladder, with one step proved-modulo-a-pin and one step *refuted*.

4.1 Gate-2: transparent composition, reduced to one pin

Corollary 8.19 lifts LKR_{in} from the single gadget to a whole cut-open radius-1 collar chain across a transparent interface. The manuscript’s own Step-1 proof of Lemma 8.18 (a pointwise preparation via Lemma 8.16) does *not* go through — the formalization refutes it directly:

```
theorem lemma818_pointwise_lift_preparation_obstruction : ... -- axiom-clean
```

In its place the development builds a frontier-automaton / fibration architecture: a finite frontier state over the chain transcript, generated parent-row certificates, and a fibered-connectivity decomposition (base connected + fibers connected + lifts). The executable chain test `chainLKRInAudit` passes on all words up to length seven (real-fiber labs, all input fibers connected, zero failures). The former metadata-only classification audit has been replaced: the actual per-target test is `representativeTargetFiberKempeOrbitConnectivityAudit`, and Gate-2's single live pin is

`Gate2RepresentativeFiberKempeOrbitConnectivityPin`

combining fourteen representative real-fiber audits with DFA-mode sufficiency; the all-chain implication is proved conditional on this pin. (A separate refuted route, `not_concreteChainFiberAppendPrefixFibrationClosed`, records a shortcut that provably fails and is kept dead.)

4.2 S2: length-2/3 radial-closing-edge fiber census

Gate-2 certifies LKR_{in} with the *first* gadget's input stubs fixed; §9.2 Step 3 needs the *cyclically re-glued* collar with all outer-side stubs fixed — different fibers, no bridging lemma in the manuscript. What is proved is a finite census: it ranges over twelve tau/mirror words of length 2 or 3 and fixes the four radial-closing edges (a lab found all 972 such fibers connected-or-empty — 216 nonempty, 756 empty, zero disconnected — and the census target was discharged with zero pins). It does *not* fix every outer stub and does not range over arbitrary collar length. The actual manuscript claim is named `ManuscriptSection92S2A11OuterStubsFixedArbitraryCollarTarget` and remains open.

```
def closedCollarLength23RadialClosingEdgeFiberCensus : Prop := ... -- proved
def ManuscriptSection92S2A11OuterStubsFixedArbitraryCollarTarget : Prop -- open
```

The route survey had rated S2 the sharpest mismatch in the manuscript — a statement never written down in v23 and not implied by Gate-2 — so proving it on the finite class is a real advance.

4.3 S4: abstract-chain ambient-escape regression

§9.2 Step 4 reinterprets a collar Kempe word as a word of the *ambient* annulus, and confines the switched components with a face-only locality lemma (7.9). The formalization proves an ambient-escape regression in the encoded tau-chain model — a collar plus one deeper radial ring sharing the interface. No theorem currently realizes that witness as a simple planar, cyclically-five-edge-connected actual collar; graph-facing salvage is conditional on `ClosedWalkActualCollarRotationFaceIncidenceCutLiftObligation`, which remains open.

```
/-- A collar-certified, collar-input-avoiding Kempe switch whose ambient
two-layer component leaves through the inner side, hits the fixed outer input,
and does NOT restrict to the certified collar move. -/
theorem section92Step4TwoLayerCounterexampleClass :
  Section92Step4TwoLayerCounterexampleClass -- axioms [proptext]
```

Concretely (outer word $\tau\tau$, inner word τ , input key r, r, b, r): a switch that is input-avoiding *in the collar alone* balloons, in the two-layer object, into a component that crosses the shared inner boundary into the deeper ring, returns, meets the fixed outer input edge, and lands the collar on a *different* coloring than the certified move. This refutes the reinterpretation as a regime-free, purely-local principle: the ambient switch is a *different* move on the collar than the one Gate-2/S2 certified.

We are precise about what this does and does not say. It is a defect in the manuscript’s assembly *seam* as argued (Step 4 invokes 7.9, a lemma proved for single-face collars, as if it were a global confinement), not a refutation of 4CT.

The cheapest repair has also been tested, and *refuted*. Strengthening the certification so that switches avoid the entire outer interface — both input *and* output stubs, not only the inputs — does eliminate the escape (a two-layer companion run finds zero escapes under full-interface avoidance), but at a fatal cost: on the same finite class, two colorings with the same fixed input are connected *only* by a switch whose component touches the output stubs, so forbidding output touches disconnects the fibers (module `GoertzelLemma818TwoSidedAvoidance`: 432 open and 216 closed nonempty fibers disconnect). The escape channel is therefore *load-bearing for connectivity*: the very moves that let a component escape the collar are the moves the fibers need to remain connected. Confinement and connectivity are in direct tension, so any repair must confine the escaping components *without* severing collar connectivity — a needle the simplest fix cannot thread.

What remains is a genuinely regime-level salvage: a proof that the minimal-counterexample *regime* (CAP5-free normal form) excludes the disconnecting/escaping configurations, or a §9 re-design that only ever peels *face* collars where 7.9 applies as stated. Neither is established; both need planar-normal-form infrastructure the development does not yet have; and the abstract gadget lab cannot decide them, since the tension may or may not dissolve in the actual normal form. As of this writing the seam is open, with both the escape and the failure of the cheapest repair pinned down as checked theorems.

4.4 Sharpening the open salvage: a winding invariant and a cyclic 2-cut reduction

Since the escape was pinned down, the formalization has *sharpened* — though not closed — the open salvage, by asking a Spencer–Brown/Kauffman-native question: which two-color components can escape? On the closed annular collar a two-color component carries a well-defined mod-2 *winding class* around the annulus generator — the parity of its crossing count with a fixed radial cut — a Parity-Lemma-style invariant, well defined per component and tracked across Kempe switches. In this language the escape is exactly a *winding* move: the components that leave through the inner side and return are those of nonzero winding class. The precise form of the S4 freedom is then a homological statement, machine-checked on the witness class: *winding is not boundary-determined* — a fixed-boundary fiber can carry components of different total winding class, so the escaping (winding-1) moves are not forced by the boundary data alone.

This localizes the difficulty to one geometric question: can a winding-freedom escape be *realized on a genuinely simple planar collar*? The formalization reduces it to a classical connectivity fact. Any simple planar realization that preserves the winding profile must replace the escape’s parallel endpoint bundle by a simple two-pole side; a tree-rank (handshaking) count — $2|\text{side edges}| = 3|\text{internal}| + 4$ forces $|\text{internal}| + 1 < |\text{side edges}|$, hence a cycle — turns that side into a *cyclic 2-cut* of the collar. A collar admitting no realized cyclic 2-cut therefore admits no escape realization:

```
-- every simple planar winding-escape realization forces a cyclic two-cut:
theorem ...Escape_forces_cyclicTwoCut_allSizes : ...
-- so a collar forbidding cyclic two-cuts (e.g. cyclically 5-edge-connected)
-- admits no escape realization:
theorem ...Escape_not_simplyRealizable_allSizes
  (h : ...ForbidsCyclicTwoCut G) : ¬ ...SimplePlanarEscapeRealization G
```

We are precise about what this buys. It is a *reduction of the open salvage to one named*

residual, not a repair: the theorem is conditional on the collar forbidding cyclic 2-cuts, and the current realization datum carries the separating-pair structure as an obligation rather than deriving it from the collar’s embedding. What it does is convert the vague “the minimal-counterexample regime should exclude the escape” into a concrete checkable target — *the escape collar carries no cyclic 2-cut*, i.e. it is cyclically sufficiently edge-connected, which is exactly the kind of property the minimal-counterexample normal form is expected to supply. And this residual is not an unformalizable abstraction: the development already proves cyclic-five-edge-connectivity for concrete graphs (and refutes it for others), so discharging or refuting it *for the escape collar* is a reachable next question — the honest current frontier of the S4 salvage.

5 The v24 route: deserts, pivots, and the necklace

Goertzel’s v24 working package (July 2026) replaces v23’s §6–7 with a new architecture: pentagonal *pivots*, hexagonal *deserts*, a *necklace* decomposition of the sphere relative to the twelve pentagons, winding and monodromy invariants along corridors, and a final *Trail Completeness* target from which the minimal-counterexample contradiction is to follow, conditional on a dependency chain L1–L10. The package proposes a universal Short Strand Conjecture and then refutes it itself with a 688-edge, reach-40 strand, so the route requires a replacement bound. This section records what the formalization has established of this route so far; the formalization targets combinatorial rotation systems throughout, not topological planarity.

Independent reproduction of the validation toolchain. Every numerical claim consumed from the package was recomputed independently before use: the annular censuses (5412 and 31704) with their Kempe fibers, the fullerene finite bases, the pentagon finite seals, tube seed growth, the desert and strand specimens (including the Short-Strand refuter), Goldberg normal-form specimens, and the GWCO exact base through twelve. A recurrent tube-ring transfer theorem replaces the package’s Pascal-triangle extrapolation (`GoertzelV24TubeRingTransfer`).

Rotation systems and facial linear algebra. A Tait-coloring layer is built over rotation systems, with each edge’s two faces identified, face simplicity reduced to two-sidedness, facial dual connectedness, and the exact facial linear algebra: facial boundary rank, primal cycle-space rank, the equality of the facial span with the cycle space, and the face winding classification (`GoertzelV24FaceBoundaryLinearAlgebra`, `GoertzelV24OrbitFaceCycleSpaceEquality`, `GoertzelV24WindingClassification`).

Curvature. The all-face discrete Gauss–Bonnet identity is proved from rotation-system data alone: the total face curvature $\sum_f (6 - \ell(f))$ equals 12 (`totalFaceCurvature_eq_twelve` in `GoertzelV24CurvatureScope`). The package’s *at-most-twelve pentagon* step, which is false as stated under minimum face size five, is corrected by a proved *weighted* pentagon budget charging faces by $6 - \ell(f)$; the fullerene hexagonal bulk is derived on orbit faces, and curvature-to-bulk corridor bounds connect the budget to corridor extraction (`GoertzelV24OrbitFaceCurvatureBulk`, `GoertzelV24BulkCorridor`).

Necklace and corridors. The locked-pentagon necklace is assembled and the singleton necklace strand forced (`GoertzelV24Necklace`). A hexagonal corridor theory is developed: defect-free corridor neighborhoods extracted from geodesic face paths; an internal rung classification with induced rung chains and the exclusion of adjacent induced

rungs; clean width-four slabs with oriented interfaces and the matching of consecutive interfaces; Tait transfer across slabs with finite color-transition alphabets and color-state words; and a sharp chirality barrier under which transfer across a chiral reversal forces the melted state (`GoertzelV24CleanHexCorridor`, `GoertzelV24HexFaceRungType`, `GoertzelV24OrientedHexSlab`, `GoertzelV24HexCorridorColorTransfer`, `GoertzelV24HexCorridorChiralityBarrier`).

Pumping. The corridor acceptance language is proved periodic and corridor pumping congruence-correct; with the recurrent tube-ring transfer these are the working tools of the Short-Strand replacement search (`GoertzelV24CorridorPumping`).

Framed trails and local fiber calculi. The framed-trail layer proves completion semantics and defect freedom — the unique admissible missing color and the characterization of proper trail extensions (`missingColorAt_is_unique_admissible`, `hasMatchingDefectColors_iff_exists_properTrailExtensionColor` in `GoertzelV24FramedTrail`) — supported by exact local calculi: the square fiber table with its realized Kempe switch and local and compositional profile identities, the digon Kempe rung, rotor local color routes, and the two-edge-cut color parity (`GoertzelV24Square/SquareKempe/SquareComposition`, `GoertzelV24Digon`, `GoertzelV24Rotor`, `GoertzelV24TwoEdgeCut`).

The minimality normal form (proved). The full minimality connectivity ladder is now machine-checked on rotation systems, with no `sorry` and axioms in `{propxt, Classical.choice, Quot.sound}`. The two-edge, cyclic three-edge, and cyclic four-edge cut reductions are each proved by capping both sides into strictly smaller cubic rotation systems and gluing their Tait colorings across the cut; a vertex-minimal Tait counterexample admits none of them, so it is cyclically five-edge-connected (`cyclicallyFiveEdgeConnected_of_vertexMinimalTaitCounterexample`). The four-edge rung is the classically hard one: at a four-edge cut the four boundary colors sum to zero in the Klein four-group, giving three boundary classes (all-equal and the two non-crossing adjacent pairings), and the two capped sides always share a reducible class. The load-bearing step — that a side admitting an all-equal boundary escapes, by a Kempe swap, into a non-crossing adjacent class (never the crossing Birkhoff pairing) — is *proved*, not assumed, from the planarity of the interior Kempe chains (`GoertzelV24FourEdgeCutKempeEscape`), and a minimal counterexample is correspondingly forced to have minimum face size five. The corridor-existence side is lifted off the fullerene premise entirely: on any spherical cubic map of minimum face size five, the pentagon count equals 12 plus the total negative curvature weight, and either the face count lies below a weight-dependent threshold or a clean hexagonal corridor exists (`GoertzelV24WeightedOrbitFaceCorridor`) — no restriction to pentagonal and hexagonal faces is used anywhere.

In progress and open. On this normal form the endgame reduction is under active construction: a fusion-chain / rebase-pumping dynamics that, from any adjacent pair of the counterexample, produces either a fusion terminal or a strictly descending rebase pumping (`exists_rotationOrderedFusionTerminalOrRebasePumping_of_vertexMinimalTaitCounterexample`), together with an exhaustive scope classification of the remote facial-dual cycles it generates (length ≥ 5 exact cut, length-three single-vertex cut, length-four adjacent-pair cut with an internal

dual chord). What remains *open* is the coloring transport that turns each classified case into a contradiction with minimality — the large-cut splice/interface law and the explicit consumption of the singleton-star and adjacent-pair-collar cases — and hence Trail Completeness, the L1–L10 chain, and a proved Short-Strand replacement. The v23-era collar obligations are taken up only where this route consumes them. Nothing in this section claims the Four-Color Theorem or the completion of the v24 route.

Canonical data at reentry joins. The rotation ordering removes a small but potentially serious compositional ambiguity. If two adjacent-pair presentations have the same ordered central edge, then their rotation-ordered four-port data are equal, regardless of the original enumeration of the four boundary ports. We therefore refine the recursive relation so that each step retains its source fusion normal and its certified pumping launch. Given two consecutive steps, the target Kempe package of the first supplies a normal with a genuine same-base Kempe escape, and the second supplies its source normal; the two recovered rotation-ordered data agree exactly. Finite iteration consequently yields a structural terminal, a large exact remote separator, or a nonempty closed orbit whose successive normal forms are canonically aligned. This alignment is a bookkeeping theorem, not yet a coloring-transport theorem: transport between different deleted adjacent pairs remains the next local collar problem.

6 Status and scope: machine-checked, reduced, and open

We state the scope exactly. The full route is a ladder $S0, \dots, S9$ (Gate-2, then v23 §9 and §3); the table records each step’s manuscript reference and current Lean status.

Step	Content	v23 ref	Lean status
—	single-gadget LKR_{in}	Lem. 8.14	finite encoded-model audit; graph move-soundness open
—	two-color transparency	Lem. 8.15	finite encoded-model audit; shares graph-fidelity bound
—	Pillar A	Thm. 4.9	actual W_0 inclusion open; unrestricted-oracle strictness open
S0	Gate-2 chain LKR_{in}	Lem. 8.18/8.19	one actual fiber/orbit plus mode-sufficiency pin
S2	radial-closing-edge fiber census	§9.2 Step 3	proved only at length 2/3; manuscript target open
S4	ambient escape regression	§9.2 Step 4	abstract-chain countermodel; graph realization open
S1	collar $\Rightarrow \tau$ /mirror chain	Lem. 8.5–8.6	open (structural)
S3	annulus peeling well-definedness	§9.2 Step 2	open (structural)
S5	interface to Kauffman Thm. 3.1	§9.2 / §3	open (statement-level)
S6	Kauffman parity/primalty internals	§3.2–3.3	open (structural)
S7	minimal-counterexample normal form	Lem. 5.2–5.3	open (structural)
S8	CAP5 reducibility	§6	partial
S9	minimality wrap (+ Tait)	Def. 5.1 / Thm. 2.3	open (structural)

Machine-checked outright. The finite encoded-model audits for Lemmas 8.14 and 8.15, together with completeness of the 192-state incidence enumeration; unrestricted-oracle strictness and conditional need-based Pillar A infrastructure (with four uniform peel oracles refuted); and the length-2/3 radial-closing-edge S2 census. The corresponding graph move-soundness, actual v23 $W_0(H)$ inclusion, and manuscript S2 statement remain open. All with no `sorry` and axioms $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$.

Reduced to one pin. Gate-2 (Lemmas 8.18/8.19): the manuscript’s pointwise Step-1 preparation is refuted and replaced by the frontier-automaton / fibration architecture, whose real-fiber labs pass to length seven; the whole composition rests on a single named structural remainder.

Regression countermodel. §9.2 Step 4’s collar-word-to-ambient-word reinterpretation fails in the encoded two-layer tau-chain model. A simple planar cyclically-five-connected graph realization has not been proved; the actual-collar cut-lift obligation remains open. In the encoded model this lowers the odds that the v23 route completes *as written*; its salvage is open.

v24 scope. See §5. The validation toolchain and censuses are independently reproduced; the all-face curvature identity and the corrected *weighted* pentagon budget are proved; the necklace/-corridor, pumping, and framed-trail layers are in the tree; the endgame (fusion chains, chiral monodromy, Trail Completeness, L1–L10, and a proved Short-Strand replacement) is open.

Open (the non-finite heart). Steps S1, S3, S5–S9: the realization of abstract chains as actual collars, the wide-annulus peeling, the interface to Kauffman’s reduction, and the Kauffman/normal-form/CAP5 internals. Their common cost is one shared planar-embedding/duality/Jordan infrastructure the development does not yet have; most carry a known proof shape, so their risk is formalization cost rather than mathematical doubt — with the exception of the S4 salvage, which is genuinely open.

What is not claimed. We do *not* claim the Four-Color Theorem, and we do not claim that v23 proves it. The 4CT is a theorem via the computer-assisted proofs; what is at issue is solely the viability of the v23 human-readable route, which remains *open* — its foundational layer now machine-checked, its most dangerous assembly seam pinned down as broken.

Provenance

The manuscript discussed is Ben Goertzel, *A Human-Readable, Spencer–Brown/Kauffman-Style Proof of the Four-Color Theorem*, version v23. The formalization is work of the MeTTa-pedia formalization effort. The keystone and transparency are module `GoertzelLemma814`; the Pillar-A obstruction and its need-based repair are `FrontierGeometric`, `GoalAudit`, and `PillarASmallerSpace`; the Gate-2 architecture is `GoertzelLemma818SemanticBridge` and the frontier-automaton modules; the S2 bridge is `GoertzelLemma818ClosedCollarS2Bridge`; the S4 counterexample and the refuted two-sided-avoidance repair are `GoertzelLemma818TwoLayerEscape` and `GoertzelLemma818TwoSidedAvoidance`. The v24 route of §5 is the `GoertzelV24*` module family (roughly sixty modules; the primary mathematical source is the v24 working package *Deserts, Pivots, and the Necklace*, July 2026). Lean listings are ASCII transliterations of the source declarations (the sources are normative); every headline was checked with `#print axioms`. Per-declaration source links are omitted pending publication of the development tree.

A A graveyard of superseded sub-routes

Each entry is a route or reading that was tried and set aside; several are themselves machine-checked theorems that a stated step *fails*.

The spanning *equality* $W_0(H) = \text{span}(\mathcal{G})$. Refuted on every closed-walk exact shell (§3); superseded by the need-based inclusion, which is what the synthesis actually consumes.

The four uniform geometric peel oracles. Each would have closed Theorem 4.9’s synthesis uniformly; all four are machine-checked false. The route uses the manuscript’s own per-embedding peeling witnesses instead.

The pointwise Step-1 preparation of Lemma 8.18. The manuscript’s own pointwise lift (via Lemma 8.16) is refuted (`lemma818_pointwise_lift_preparation_obstruction`); replaced by the frontier-automaton / fibration composition.

The prefix-fibration shortcut for Gate-2. A tempting closed-form shortcut, machine-checked to fail (`not_concreteChainFiberAppendPrefixFibrationClosed`); kept dead.

The false “additivity” of Kempe switches. An $\mathbb{F}_2$ -additive execution of the Pillar-A decomposition $\Delta = \sum v_{f,g}$. This is *not* the manuscript’s route — Pillar C is titled “executability without false additivity,” and v23 §8.1–8.3 show mixed color-pair switches on shared support do not commute. Recorded only to set aside; the faithful route is the operator/LKR_{in} execution of §2.

An $\mathbb{F}_2$ -additivity / evader reconstruction (disavowed). An early exploratory reconstruction outside the manuscript’s route, which v23 explicitly disavows (additivity invalid for mixed color-pairs; Rmk. 8.10). Not part of this companion’s formalized content.